

3D NOR-type FeFETs with Record Endurance of 10^{11} , Fast Erase of 50 ns, and Immediate Read-After-Write for In-Memory Learning

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Abstract

In this work, we experimentally demonstrate a 3D NOR-type FeFET (FeNOR) array based on ALD IGO channel. With a scaled channel area down to $0.007 \mu\text{m}^2$, the 3D FeNOR achieves a record endurance of 10^{11} cycles, 50 ns erase speed, immediate read-after-write, and minimal layer-to-layer variations. Comprehensive analysis unveils the mechanism of endurance degradation and optimization strategies. Based on multi-domain TCAD simulations, the improvement of erase speed is well-fitted with experiments and mainly attributed to blocking of percolation paths. The good density, reliability and speed make the 3D FeNOR promising for next-generation AI applications, especially in-memory learning.

Introduction

In-memory learning is a key enabler for advanced artificial intelligence (AI) with efficient data processing capability [1]. This places higher demands on memory performance, including density, speed, endurance, and power consumption (Fig.1). 3D NOR-type FeFET (FeNOR) is a prominent candidate to meet these requirements. The 3D stacked NOR array provides high density and random accessible memory (RAM), while the non-volatile and field-driven polarization switching enable low power operation (Fig.1(c) and (d)). However, existing 3D NOR and FeNOR memory still have limited endurance $< 10^8$ [2], and thus unable to support frequent writing. In addition, typical high-endurance strategy based on oxide semiconductor (OS) channels brings erase issues, with $>1 \mu\text{s}$ erase time [3]. In this work, we experimentally demonstrate a 3D FeNOR array with record endurance of $>10^{11}$ cycles, fast erase speed of 50 ns, immediate read-after-write, and low variations in scaled devices. A combination of electrical measurements, material characterizations, and modeling provides insight into the mechanisms of endurance and erase speed improvement.

Fabrication and Characterization of 3D FeNOR

The 3D FeNOR is fabricated following the process flow in Fig. 2(d). $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ (HZO) and IGO are grown by atomic layer deposition (ALD) as the FE layer and channel, respectively. The SEM image in Fig. 2(a) shows the $8 \times 8 \times 3$ 3D FeNOR array, and the shortest channel length (L_{ch}) is 100 nm (Fig.2 (b)). The cross-sectional HRTEM images in Fig. 2(e) and (f) show the uniform three-layer 3D stacked structure and the clear interface between the HZO and the IGO.

The $I_{\text{d}}\text{-}V_{\text{g}}$ sweep for the 3D FeNORs with different L_{ch} are displayed in Fig. 3(a), showing $I_{\text{on}}/I_{\text{off}} > 10^5$. All FeNORs exhibit typical counter-clockwise hysteresis, and the devices with the smallest L_{ch} have the largest memory window (MW) of ~ 1.0 V. Devices in the three layers of the same trench show nearly identical $I_{\text{d}}\text{-}V_{\text{g}}$ curves, demonstrating excellent device uniformity (Fig. 3(b)). The MW reaches 0.7 V after programming and erasing with pulses, as shown in Fig.3(c). Pulses with different amplitudes and widths are applied to determine the write speed of the 3D FeNOR, as illustrated in Fig. 4(a) and (b). Fast program and erase speed of 50 ns are achieved. Note that this erase speed is much faster than most of the oxide channel FeFETs. The read-after-write-delay (RAW) of the 3D FeNOR is characterized in Fig. 4(c), with read latency of <100 ns.

Endurance measurement for the 3D FeNOR is depicted in Fig. 5(a). With write pulses of $+4.25$ V/ -3.25 V, 50 ns, MW reaches 0.5 V initially and remains >0.3 V after 10^{11} bipolar cycles. Fig. 5(b) shows the read $I_{\text{d}}\text{-}V_{\text{g}}$ curves with cycling, with slight MW decrease. Retention is measured at room

temperature and 85°C , both demonstrating stability over 4000 s and extrapolated to >10 years (Fig.5(c)).

Investigation of Endurance and Fast erase

A series of tests are then conducted to investigate the factors affecting endurance. Distinct pulse conditions with equal initial MW are first studied in Fig. 6(a). Pulses with shorter widths effectively mitigate MW degradation, suggesting suppression of charge trapping during writing. The three layers of FeNOR in the same trench exhibit virtually identical endurance processes (Fig.6(b)). FeNOR with a shorter L_{ch} exhibits better endurance, mainly attributed to the smaller erase voltage (Fig.6(c)). Electron Energy Loss Spectroscopy (EELS) is applied to compare the pristine and cycled cells (Fig.7(a)), and results are shown in Fig.7(b). The B/A peak ratio in O-K spectroscopy signifies the concentration of oxygen vacancy (V_{o}) [4]. The V_{o} show an increasing gradient from HZO to IGO in pristine cells, while they are uniformly distributed across the interface in cycled cells. This indicates the migration of V_{o} from the IGO to the HZO interface during cycling. The precession electron diffraction (PED) measurement in Fig. 7(c) exhibits large amount of FE o-phase in the cycled cells, with no significant FE fatigue. Fig. 7(d) illustrates the schematic of endurance degradation, suggesting reducing V_{o} in oxide channel surface as a possible method to further enhance the endurance.

To systematically study the geometric factors related to write speed, planer IGO FeFETs with various channel widths and lengths are fabricated (Fig. 8). The program and erase speed are plotted in Fig. 9, showing faster erase speed at smaller widths and lengths. To better understand the size effect, device modeling is constructed with the framework and the key parameters in Fig. 10 (a) and (b). Fig. 10 (c) and (d) show the erasing process of FeFET. As polarization gradually switches, the electron density of each region decreases and shift the $I_{\text{d}}\text{-}V_{\text{g}}$ curve to the positive direction. The simulated write speed matches with measurement results (Fig. 10(e)). The effect of channel width and length on erase speed is illustrated in Fig. 11. The absence of percolation paths in narrower channel and the stronger fringing field effect in shorter channel lead to faster erase speed.

Simulation and Benchmark

Fig. 12(a) depicts the architecture of a multi-bit high-bandwidth digital CIM based on 3D FeNOR. The measured logic operation for the 1-bit FeNOR cell is displayed in Fig. 12 (b), showing correct results. The MNIST task is simulated based on the extracted device characteristics (Fig. 12(c)). The recognition accuracy of 3D FeNOR is comparable to software, and shows little degradation even after 10^{11} cycles. Finally, the key memory performance of our 3D FeNOR is benchmarked with other 3D transistor memory in Table. 1 [2,5-8]. The fabricated 3D FeNOR makes great breakthroughs in endurance, and shows superiority in speed, operation voltage, and RAW.

Conclusion

In this work, we experimentally demonstrate a 3D FeNOR with record endurance of 10^{11} , fast erase speed of 50 ns, immediate read-after-write, and good uniformity in scaled devices. Comprehensive analysis is conducted to reveal the mechanisms for endurance and erase speed improvement. This work facilitates development of high-density, reliable, and energy-efficient memory for future AI with learning capability.

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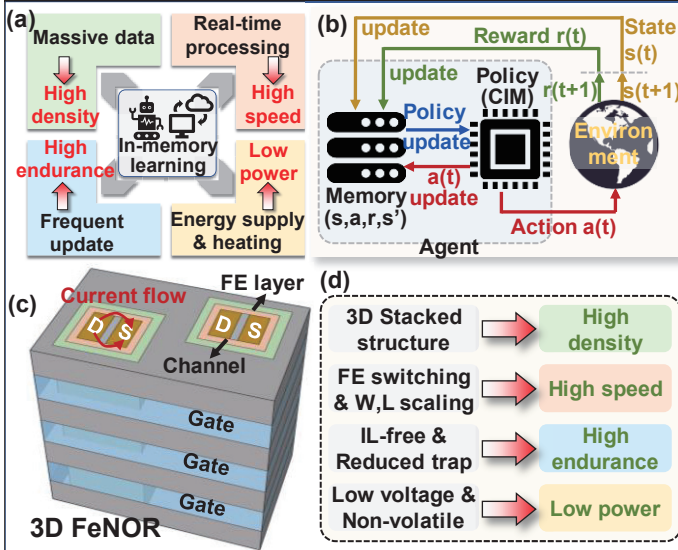


Fig. 1 (a) Demand for memory by in-memory learning. High density/speed, low power and non-volatility are required. (b) Schematic of in-memory reinforcement learning (RL) system. (c) Structure and (d) advantages for 3D FeNOR, which perfectly fulfills the demands of in-memory learning.

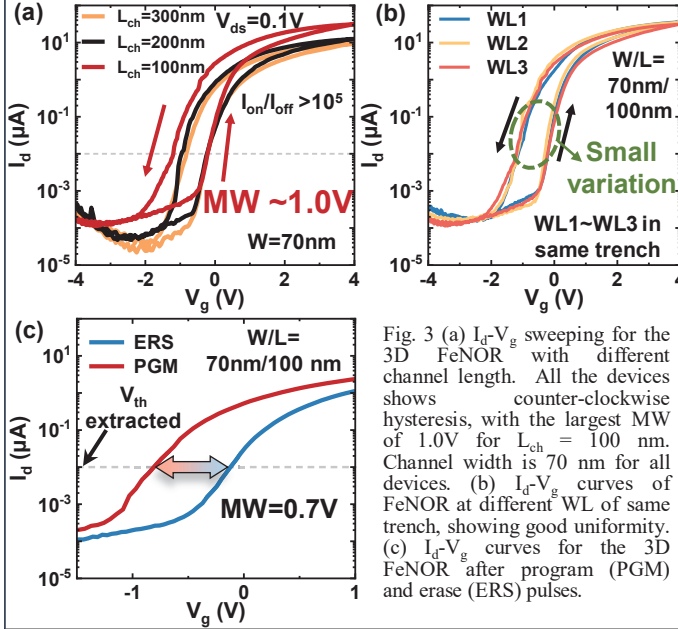


Fig. 3 (a) I_d - V_g sweeping for the 3D FeNOR with different channel length. All the devices shows counter-clockwise hysteresis, with the largest MW of 1.0V for $L_{ch} = 100nm$. Channel width is 70 nm for all devices. (b) I_d - V_g curves of FeNOR at different WL of same trench, showing good uniformity. (c) I_d - V_g curves for the 3D FeNOR after program (PGM) and erase (ERS) pulses.

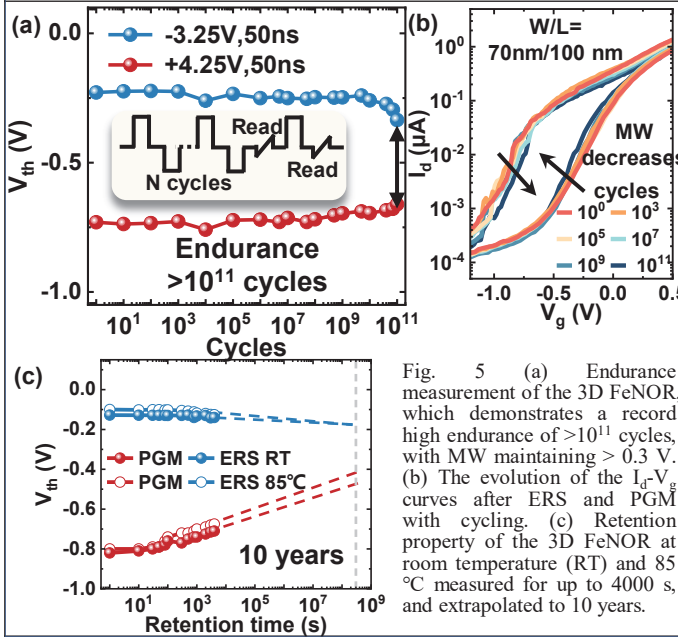


Fig. 5 (a) Endurance measurement of the 3D FeNOR, which demonstrates a record high endurance of $> 10^{11}$ cycles, with MW maintaining $> 0.3V$. (b) The evolution of the I_d - V_g curves after ERS and PGM with cycling. (c) Retention property of the 3D FeNOR at room temperature (RT) and 85 °C measured for up to 4000 s, and extrapolated to 10 years.

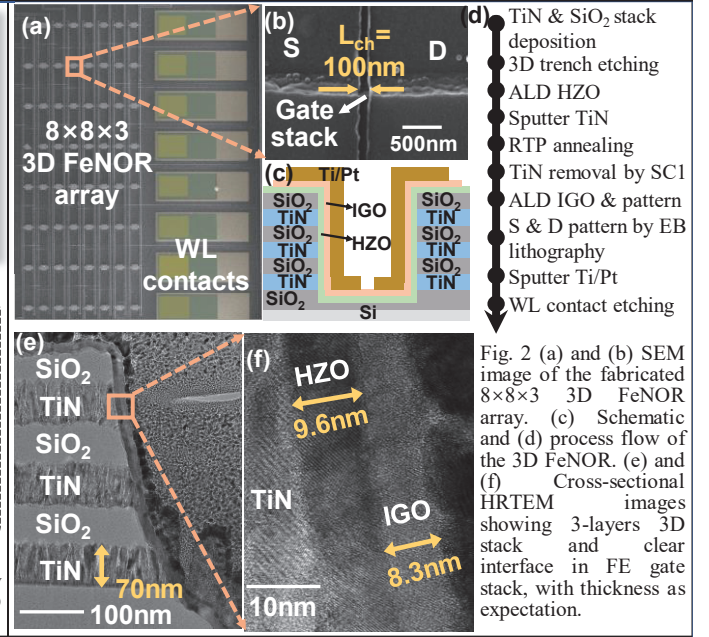


Fig. 2 (a) and (b) SEM image of the fabricated 8x8x3 3D FeNOR array. (c) Schematic and (d) process flow of the 3D FeNOR. (e) and (f) Cross-sectional HRTEM images showing 3-layers 3D stack and clear interface in FE gate stack, with thickness as expectation.

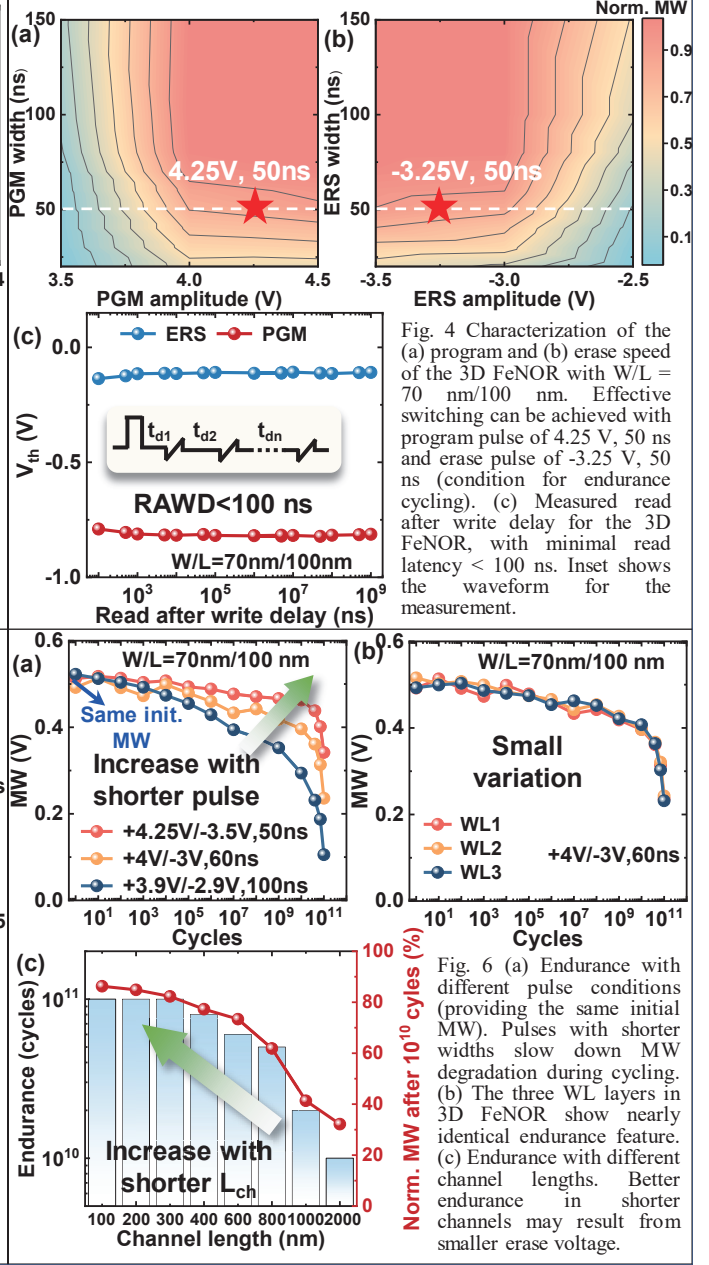


Fig. 6 (a) Endurance with different pulse conditions (providing the same initial MW). Pulses with shorter widths slow down MW degradation during cycling. (b) The three WL layers in 3D FeNOR show nearly identical endurance feature. (c) Endurance with different channel lengths. Better endurance in shorter channels may result from smaller erase voltage.

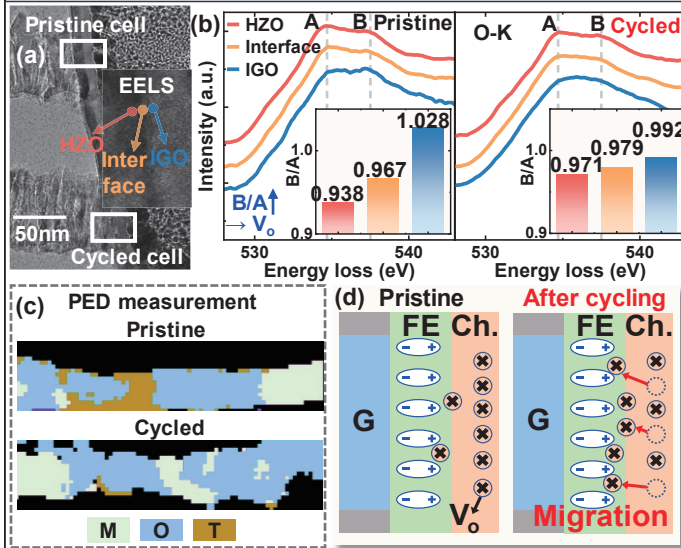


Fig. 7 (a) and (b) O-K EELS measurement for pristine and cycled cells, indicating V_o migrating from IGO to HZO during cycling. (c) PED measurement for the HZO layer, showing plentiful o-phase for cycled cells. (d) Schematic of endurance degradation mechanism.

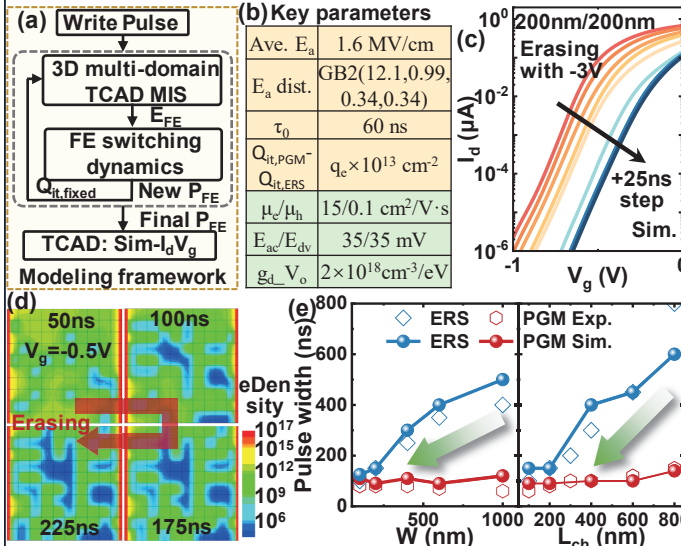


Fig. 10 (a) The framework and (b) key parameters of the modeling in Sentaurus TCAD. (c) Simulated I_d - V_g curves and (d) corresponding eDensity in IGO channel with increasing erase pulse width. (e) Simulated program and erase speed with various channel widths and lengths.

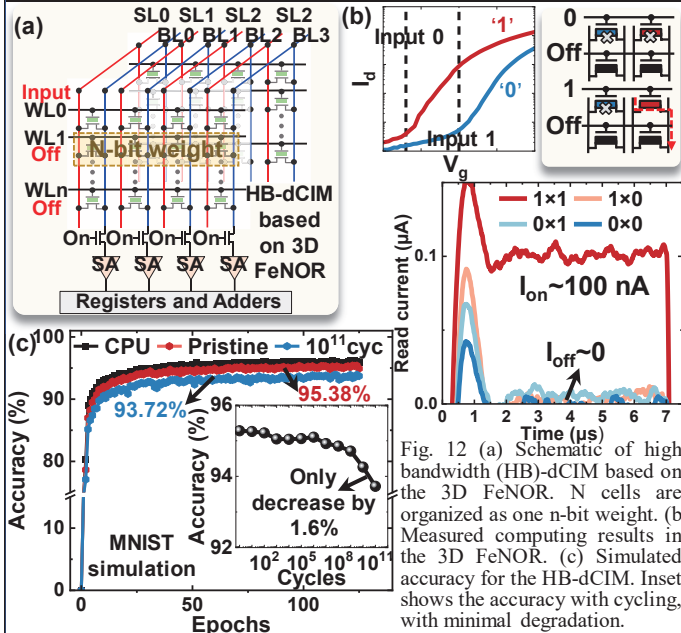


Fig. 12 (a) Schematic of high bandwidth (HB)-dCIM based on the 3D FeNOR. N cells are organized as one n-bit weight. (b) Measured computing results in the 3D FeNOR. (c) Simulated accuracy for the HB-dCIM. Inset shows the accuracy with cycling, with minimal degradation.

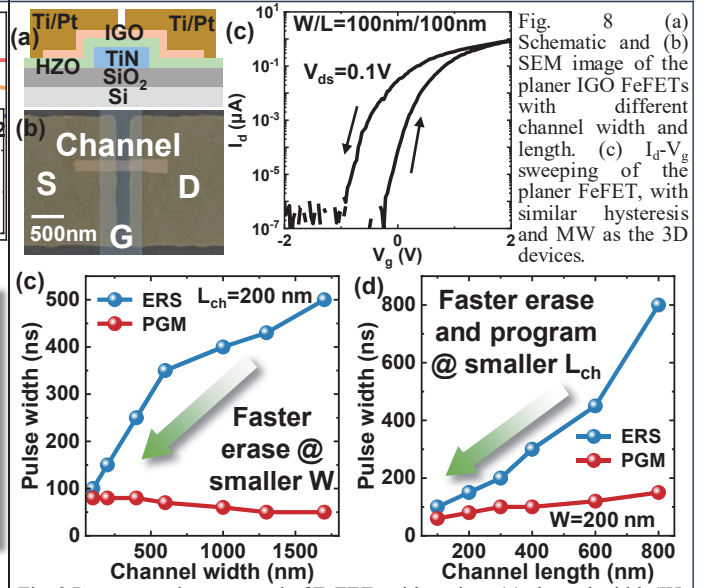


Fig. 9 Program and erase speed of FeFETs with various (a) channel width (W) and (b) channel length (L_{ch}). The erase speed is faster at smaller W and L_{ch} .

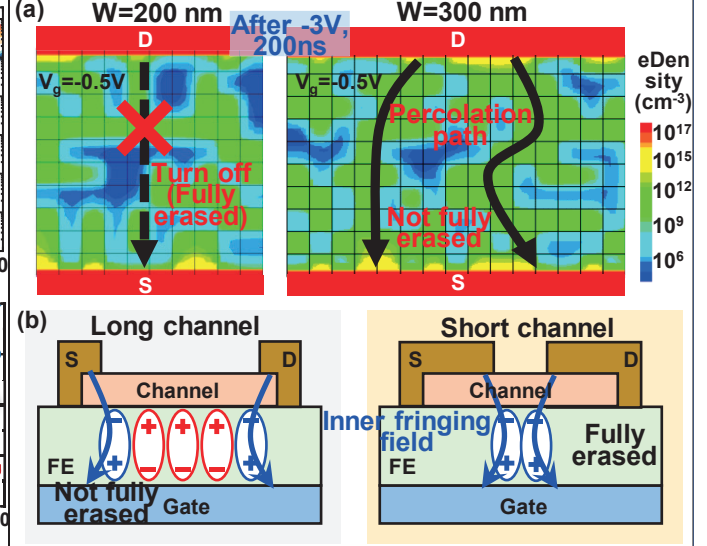


Fig. 11 Mechanisms for the faster erase speed at (a) smaller channel width and (b) smaller channel length. The absence of current percolation path in wider channel and the greater effect of inner fringing field in shorter channels lead to fully erased state.

Table 1. Benchmark of the memory performance for our 3D FeNOR with other 3D transistor memory

	NC23 [5]	VLSI21 [6]	EDL22 [7]	VLSI24 [8]	VLSI24 [2]	This work
Structure	3D FeNAND	3D NOR flash	3D NOR flash	Vertical FE-CAA	3D FeNOR	3D FeNOR
Channel	IZO	Poly-Si	Si	TiO ₂	IZO	IGO
PGM condition	4V	24V	10V	-	5V	4.25V
	10ms	1μs	10μs	-	25μs	50ns
ERS condition	-5V	-17V	-12V	-	-5V	-3.25V
	10ms	5μs	100ms	-	25μs	50ns
RAWD	-	-	-	-	-	<100ns
MW	1.5V	4.5V	1.2V	~1.0V	4.0V	1.0V
Endurance	10 ⁶	5×10 ⁷	10 ⁵	10 ⁶	10 ⁷	>10 ¹¹

Reference [1] Kim, J. *et al.*, *Adv. Intell. Syst.*, 6: 2300763, 2024. [2] Y. Feng *et al.*, *Symp. on VLSI*, 2023, pp.1-2. [3] Y. -H. Chen *et al.*, *EDTM*, 2023, pp. 1-3. [4] J. Chen *et al.*, *IEDM*, 2023, pp. 1-4. [5] Kim, I. J. *et al.*, *Nat. Com.*, vol. 14, no. 1, pp. 504, 2023. [6] H. T. Lue *et al.*, *Symp. on VLSI*, 2021, pp. 1-2. [7] W. Huang *et al.*, *IEEE EDL*, vol. 43, no. 11, pp. 1874-1877, 2022. [8] S. Kabuyanagi *et al.*, *Symp. on VLSI*, 2024, pp. 1-2.